

KUMAGAYA, Japan

WMO: 476260

Lat: 36.1503N

Lon: 139.3806E

Elev: 104

StdP: 14.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 28.1	(c) 29.9	(d) 7.2	(e) 8.0	(f) 42.7	(g) 10.1	(h) 9.3	(i) 41.6	(j) 20.7	(k) 44.8	(l) 18.9	(m) 44.6	(n) 4.6	(o) 290	(p) 0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 15.0	(c) 96.2	(d) 77.9	(e) 93.6	(f) 77.4	(g) 90.9	(h) 76.2	(i) 79.8	(j) 91.0	(k) 78.9	(l) 89.2	(m) 77.9	(n) 87.2	(o) 7.5	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.2	141.9	83.6	76.3	137.6	83.2	75.3	133.3	82.9	43.5	91.2	42.5	89.4	41.5	87.5	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	15.3	13.0		24.4	100.5	1.2	2.0	23.5	101.9	22.8	103.0	22.1	104.1	21.2	105.6
				21.3	81.5	1.3	1.0	20.3	82.2	19.6	82.8	18.9	83.4	17.9	84.2
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.6	40.3	41.7	47.9	57.7	66.8	72.7	80.1	82.2	74.9	64.4	53.4	44.4
	DBStd	15.36	3.73	4.49	5.72	6.18	5.14	5.02	5.77	5.06	5.87	5.28	5.19	4.59
	HDD50	870	303	237	113	11	0	0	0	0	0	0	23	183
	HDD65	3288	767	653	532	228	41	4	0	0	3	74	347	639
	CDD50	4754	1	4	46	241	519	681	934	998	746	448	126	10
	CDD65	1697	0	0	0	9	96	235	469	533	299	56	0	0
	CDH74	15006	0	0	3	94	671	1526	4610	5705	2168	227	2	0
	CDH80	6016	0	0	0	11	149	460	2048	2575	736	37	0	0
	WSAvg	5.5	6.6	6.9	6.8	6.1	5.5	5.0	4.8	4.8	4.6	4.6	4.9	5.9
	PrecAvg	54.1	1.6	1.5	2.8	3.7	4.8	6.0	6.9	7.5	8.6	6.5	2.6	1.5
(15) Precipitation	PrecMax	73.9	4.3	4.8	5.1	8.8	9.4	15.2	13.1	17.0	16.5	20.4	10.6	6.1
	PrecMin	39.8	0.1	0.2	1.0	0.9	1.5	2.0	2.8	1.2	1.1	0.7	0.1	0.2
	PrecStd	7.8	1.3	1.1	1.1	2.2	2.0	2.6	3.0	4.5	3.8	4.8	2.0	1.3
	PrecStd	7.8	1.3	1.1	1.1	2.2	2.0	2.6	3.0	4.5	3.8	4.8	2.0	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	58.9	64.5	72.5	81.1	88.3	92.5	99.0	99.2	94.2	84.4	72.0	63.5	54.0
	0.4% MCWB	45.5	50.7	53.6	60.6	64.9	73.2	78.0	78.5	76.3	68.7	57.7	51.3	44.0
	2% DB	54.1	57.8	66.8	76.5	83.7	88.2	96.2	96.7	90.8	79.5	68.6	59.0	51.3
	2% MCWB	41.9	44.8	51.2	57.9	63.7	70.8	78.0	78.2	75.0	65.9	56.3	47.7	41.3
	5% DB	51.2	54.2	62.7	72.9	80.8	85.2	93.5	94.3	87.5	76.1	65.5	56.1	47.4
	5% MCWB	40.1	42.6	48.8	56.9	62.8	69.5	77.4	77.9	74.3	64.1	54.3	45.6	41.3
	10% DB	48.8	51.0	58.8	69.4	77.7	82.2	90.4	91.7	84.2	73.3	62.8	53.3	45.6
	10% MCWB	38.7	40.4	47.5	55.6	62.2	68.7	76.4	77.4	72.4	62.4	53.4	44.0	41.3
	0.4% WB	47.6	52.7	57.3	67.1	71.1	77.1	81.0	81.3	79.0	71.9	62.2	54.0	47.4
	0.4% MCDB	52.8	62.0	65.2	73.1	79.1	86.4	93.4	93.3	88.0	78.7	67.4	58.9	51.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	44.0	47.1	54.2	62.6	68.0	74.9	79.6	80.2	77.4	69.4	59.8	50.0	47.4
	2% MCDB	52.0	55.1	62.3	71.3	77.1	83.7	91.3	91.5	85.1	75.6	64.5	55.9	47.4
	5% WB	41.4	44.2	51.4	60.1	66.2	73.0	78.6	79.2	76.4	67.3	57.3	47.4	41.3
	5% MCDB	49.3	51.5	60.0	68.6	74.8	80.8	89.5	89.5	83.5	73.0	61.9	54.0	41.3
	10% WB	39.5	41.7	48.5	57.9	64.6	71.2	77.6	78.4	75.0	65.2	55.2	45.1	41.3
	10% MCDB	47.3	49.3	57.3	65.6	72.7	77.9	87.1	87.8	81.6	70.9	60.5	51.6	41.3
(35) Mean Daily Temperature Range	MDBR	17.1	17.2	18.0	18.9	17.8	14.5	14.6	15.0	13.7	14.1	16.0	16.9	16.9
	5% DB	20.6	22.5	24.1	24.7	23.7	20.1	19.7	19.2	18.0	17.8	19.2	20.0	20.0
	MCWBR	12.0	13.1	13.3	11.8	9.6	7.4	6.2	5.6	6.2	8.1	10.3	11.6	11.6
	5% WB	19.3	20.4	21.0	20.0	18.8	15.7	17.9	17.3	15.6	14.3	14.9	17.8	17.8
	MCWBR	12.0	13.1	13.2	11.7	9.2	6.6	6.3	5.8	6.0	7.6	9.4	11.4	11.4
	MCWBR	12.0	13.1	13.2	11.7	9.2	6.6	6.3	5.8	6.0	7.6	9.4	11.4	11.4
(40) Clear-Sky Solar Irradiance	taub	0.365	0.411	0.481	0.538	0.570	0.616	0.608	0.605	0.525	0.477	0.410	0.368	0.368
	taud	2.242	2.105	1.948	1.868	1.847	1.802	1.873	1.869	2.042	2.099	2.214	2.278	2.278
	Ebn at Noon	256	256	248	241	235	224	225	222	233	232	238	247	247
	Edh at Noon	34	43	55	63	65	69	63	62	50	43	35	30	30
(44) All-Sky Solar Radiation	RadAvg	846	1047	1310	1522	1638	1455	1493	1491	1191	977	805	744	744
	RadStd	51	81	77	135	168	122	238	154	102	98	70	51	51

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page